Physics Reference Notes

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1 Similarity transformations of operators: computing $e^A B e^{-A}$

1.1 Motivation

Similarity transformations $B \mapsto e^A B e^{-A}$ are ubiquitous whenever a Hamiltonian is brought to a simpler (often diagonal) form: Bogoliubov transformations of quadratic Hamiltonians, displacement/squeezing of bosonic modes, polaron (Lang-Firsov) transformations, Schrieffer-Wolff transformations, and passage to the interaction picture or a rotating frame are all instances of this construction. If $A^\dagger = -A$ (generator anti-Hermitian), $U = e^A$ is unitary and the transformation preserves Hermiticity and spectra. This is the case in all examples below.

1.2 Method A: The Hadamard lemma

Hadamard Lemma. *For any two operators A, B,*

$$e^A B e^{-A} = \sum_{n=0}^{\infty} \frac{1}{n!} \operatorname{ad}_A^n(B), \qquad \textit{with} \tag{1}$$

$$\operatorname{ad}_A^n(B) \equiv [A,B]_n \equiv [A,[A,B]_{n-1}], \quad \operatorname{ad}_A^0(B) \equiv [A,B]. \tag{2}$$

This series is especially helpful whenever the nested commutator yields zero at for a finite n or if the nested commutators eventually reproduce a small, recognizable pattern.

Strategy 1: Reduce to elementary operators. Utilize that for any A, X, Y,

$$e^A (XY) e^{-A} = \big(e^A X e^{-A}\big)\big(e^A Y e^{-A}\big). \tag{3}$$

It thus suffices to compute $e^A\, a\, e^{-A}$ and $e^A\, a^\dagger e^{-A}$ once and any polynomial in these follows by substitution and normal-ordering, with no further Hadamard series needed.

Strategy 2: The dagger shortcut. If $A^\dagger = -A$, $U = e^A$ is unitary with $U^\dagger = e^{-A} = U^{-1}$, so $e^A B e^{-A} = U B U^\dagger$. Therefore, taking the dagger simply yields

$$\big(e^A B e^{-A}\big)^\dagger = e^A B^\dagger e^{-A}. \tag{4}$$

Once $e^A a e^{-A}$ is known, $e^A a^\dagger e^{-A}$ follows for free by Hermitian conjugation, roughly halving the bookkeeping.

1.3 Method B: The generator flow method (not proof read since not used yet)

Define $X(\theta) \equiv e^{\theta A}\, X\, e^{-\theta A}$, $X(0) = X$. Differentiating,

$$\frac{dX(\theta)}{d\theta} = [A, X(\theta)] = \operatorname{ad}_A\big(X(\theta)\big). \tag{5}$$

This ordinary differential equation (ODE) is exactly equivalent to Method A – the Hadamard series is its formal Taylor solution around $\theta = 0$ – but is can be more tractable in practice.

Key fact: closure $\Rightarrow$ finite-dimensional linear ODE. If ad_A maps a finite set of operators $\{X_1, \ldots, X_n\}$ into linear combinations of themselves, the operator ODE collapses to a matrix ODE:

$$\frac{d}{d\theta}\vec{X}(\theta) = M\,\vec{X}(\theta), \qquad [A, X_j] = \sum_i M_{ij} X_i, \qquad \vec{X}(\theta) = e^{\theta M}\vec{X}(0), \tag{6}$$

reducing the problem to exponentiating an ordinary $n \times n$ matrix.

1.4 When to prefer which method

Both methods are mathematically identical. Method A is fast when the pattern is recognizable by eye; Method B is more systematic for anything beyond the simplest single-operator case, especially with mixed or multiple generators to track jointly.

1.5 Scope: when this machinery fails

The closure property requires A to be at most quadratic in a set of canonically (anti-)commuting operators. For a genuinely quartic operator such as $H_{\mathrm{int}} = U\sum_i (n_i - \frac{1}{2})(n_{i+1} - \frac{1}{2})$, $\mathrm{ad}_A(H_{\mathrm{int}})$ does not close on a finite set for generic one-body A (e.g. $A \propto J$ or $A \propto T$) – $[J, H_{\mathrm{int}}]$ is generically nonzero and does not reduce to a small closed set of operators. The Hadamard series then cannot be resummed in closed form; one is limited to a perturbative treatment in the strength of A.

1.6 References

- J.J. Sakurai & J. Napolitano, *Modern Quantum Mechanics.*
- C. Gerry & P. Knight, *Introductory Quantum Optics.*
- D.F. Walls & G.J. Milburn, *Quantum Optics.*
- J.-P. Blaizot & G. Ripka, *Quantum Theory of Finite Systems*, MIT Press.
- A. Perelomov, *Generalized Coherent States and Their Applications*, Springer.
- J.W. Negele & H. Orland, *Quantum Many-Particle Systems.*
- B.C. Hall, *Lie Groups, Lie Algebras, and Representations.*

2 The tensor product and the Kronecker product as one of its possible representations

2.1 The tensor product of vector spaces

Let $\underline{v} = \{v_i \mid i \in I\}$ and $\underline{w} = \{w_j \mid j \in J\}$ be bases for V and W. Then $V \otimes W$ is the product space, where there exists a basis which can be identified uniquely with the pairs of the cartesian product

$$\underline{v} \times \underline{w} = \big\{(v_i, w_j) \mid i \in I,\ j \in J\big\}, \tag{7}$$

so that

$$\dim(V \otimes W) = \dim V \cdot \dim W. \tag{8}$$

Since $V \otimes W$ is a vector space, an arbitrary element of it can be written as

$$\sum_{(i,j)\in I\times J} c_{ij}\,(v_i \otimes w_j). \tag{9}$$

The bilinear map $\otimes$ is defined by

$$\otimes\colon V \times W \longrightarrow V \otimes W, \tag{10}$$

$$(v,w) \longmapsto v \otimes w = \sum_{(i,j)\in I\times J} a_i b_j\,(v_i \otimes w_j), \tag{11}$$

for $v = \sum_i a_i v_i$ and $w = \sum_j b_j w_j$. As a check of bilinearity, writing $v' = \sum_i a_i' v_i$ and $v'' = \sum_i a_i'' v_i$,

$$\begin{aligned}
(v' + v'') \otimes w &= \Big(\textstyle\sum_i a_i' v_i + \sum_i a_i'' v_i\Big) \otimes w = \Big(\textstyle\sum_i (a_i' + a_i'')\, v_i\Big) \otimes w \\
&= \sum_{i,j} (a_i' + a_i'')\, b_j\,(v_i \otimes w_j) \\
&= \sum_{i,j} a_i' b_j\,(v_i \otimes w_j) + \sum_{i,j} a_i'' b_j\,(v_i \otimes w_j) \\
&= v' \otimes w + v'' \otimes w,
\end{aligned} \tag{12}$$

and analogously in the second argument.

2.2 The tensor product of operators

Introduce operators $A\colon V \to V$ and $B\colon W \to W$ with V, W vector spaces. How is $A \otimes B\colon V \otimes W \to V \otimes W$ given? Define a map $\beta\colon V \times W \to V \otimes W$, $\beta(v,w) := Av \otimes Bw$. It is bilinear, e.g.

$$\begin{aligned}
\beta(v_1 + v_2, w) &= A(v_1 + v_2) \otimes Bw = (Av_1 + Av_2) \otimes Bw \\
&\overset{\otimes \text{ bilin.}}{=} Av_1 \otimes Bw + Av_2 \otimes Bw = \beta(v_1, w) + \beta(v_2, w),
\end{aligned} \tag{13}$$

and analogously in the second argument. By the universal property there exists a unique linear $\tilde{\beta}$ with $\tilde{\beta}(v \otimes w) := \beta(v,w)$, so that

$$(A \otimes B)(v \otimes w) = Av \otimes Bw. \tag{14}$$

2.3 The Kronecker product as a matrix representation

Let $V = K^m$ and $W = K^n$ and identify (this is *not* a unique representation!) $K^m \times K^n \to K^{(m \cdot n)}$ with the Kronecker product, i.e. for

$$v = \begin{pmatrix} a_1 \\ \vdots \\ a_m \end{pmatrix} \quad \& \quad w = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \qquad \text{set} \qquad v \otimes w := \begin{pmatrix} a_1\, w \\ \vdots \\ a_m\, w \end{pmatrix} = \begin{pmatrix} a_1 b_1 \\ \vdots \\ a_1 b_n \\ \vdots \\ a_m b_1 \\ \vdots \\ a_m b_n \end{pmatrix}. \tag{15}$$

Now choose $\{v_i\}$ to be the standard basis, so that $Av_i = A_{\cdot i}$ (the i-th column of A) and likewise for B. Then

$$\begin{aligned}(A \otimes B)(v_i \otimes v_j) = (Av_i) \otimes (Bv_j) &\overset{\text{std. basis}}{=} A_{\cdot i} \otimes B_{\cdot j} \\ = \begin{pmatrix} A_{1i}\, B_{\cdot j} \\ \vdots \\ A_{mi}\, B_{\cdot j} \end{pmatrix} &= \begin{pmatrix} \begin{pmatrix} A_{1i} B_{1j} \\ \vdots \\ A_{1i} B_{nj} \end{pmatrix} \\ \vdots \\ \begin{pmatrix} A_{mi} B_{1j} \\ \vdots \\ A_{mi} B_{nj} \end{pmatrix} \end{pmatrix}.\end{aligned} \tag{16}$$

Setting $v_i \otimes v_j \equiv w_k$ with $k = j + n\, i$ (for zero-based i), the matrix of $A \otimes B$ is the block matrix

$$\begin{aligned} A \otimes B &= \begin{pmatrix} A_{11} B_{\cdot 1} & \cdots & A_{11} B_{\cdot n} & \cdots & A_{1m} B_{\cdot 1} & \cdots & A_{1m} B_{\cdot n} \\ \vdots & & & & & & \vdots \\ A_{m1} B_{\cdot 1} & \cdots & A_{m1} B_{\cdot n} & \cdots & A_{mm} B_{\cdot 1} & \cdots & A_{mm} B_{\cdot n} \end{pmatrix} \\ &= \begin{pmatrix} A_{11} B & \cdots & A_{1m} B \\ \vdots & \ddots & \vdots \\ A_{m1} B & \cdots & A_{mm} B \end{pmatrix}, \end{aligned} \tag{17}$$

which is the Kronecker product for matrices.

2.4 References

- *Tensorprodukt* and *Kronecker-Produkt*, German Wikipedia.

3 The Jordan–Wigner mapping for spinless Fermions

3.1 Summary

The Jordan–Wigner transformation (JWT) can be used to map a fermionic system to a system of spins. This is essentially done by rewriting

$$\text{fermion} = \text{string} \times \text{operator},$$

where the "string" is some product of operators $\hat{Z}_i = 1 - 2\hat{n}_i$ and the "operator" is found to obey spin commutation relations.

As references I used

- *Fermions and Jordan–Wigner String* – ITensor C++ Documentation,
- *Introduction to Many-Body Physics*, Chapter 4.1 – Piers Coleman,
- *The Fermionic canonical commutation relations and the Jordan–Wigner transform* – M. Nielsen,

which I supplemented with some calculations and my own way of structuring things.

3.2 Setup: the fermionic Hilbert space

Our fermionic Hilbert space is defined by (Nielsen):

i) operators $c_1, \ldots, c_L$ acting on some Hilbert space;

ii) these operators satisfying the canonical (anti-)commutation relations for fermions

$$\{c_i^{(+)}, c_j^{(+)}\} = 0 \qquad \& \qquad \{c_i, c_j^\dagger\} = \delta_{ij}\, \mathbb{1}. \tag{18}$$

Many things directly follow from this, such as a possible basis for the Hilbert space, namely the occupation number basis

$$|\vec{n}\rangle = |n_1, n_2, \ldots, n_L\rangle \equiv \left(\prod_{i=1}^{L} (c_i^\dagger)^{n_i} \right) |0\rangle\,, \tag{19}$$

where $\hat{n}_i\,|\vec{n}\rangle = c_i^\dagger c_i\,|\vec{n}\rangle = n_i\,|\vec{n}\rangle$ with $n_i = 0, 1$ and

$$[\hat{n}_i, \hat{n}_j] = 0. \tag{20}$$

3.3 The transformation

The JWT introduces new operators a_j by writing

$$c_j \equiv \left(\prod_{i=1}^{j-1} Z_i\right) a_j \qquad \& \qquad c_j^\dagger \equiv a_j^\dagger \left(\prod_{i=1}^{j-1} Z_i\right), \tag{21}$$

with $Z_j = (-1)^{n_j} = 1 - 2n_j$ and $n_j = c_j^\dagger c_j$. The ordering of the Z_i does not matter since $[Z_i, Z_j] = 0$ due to eq. (20). Inverting eq. (21) using $Z_i^2 = \mathbb{1}$ and $[Z_i, Z_j] = 0$ yields

$$a_j = \left(\prod_{i=1}^{j-1} Z_i\right) c_j \qquad \& \qquad a_j^\dagger = c_j^\dagger \left(\prod_{i=1}^{j-1} Z_i\right). \tag{22}$$

3.4 Algebra of the new operators

One can show that these new operators obey

$$[a_i^{(+)}, a_j^{(+)}] = 0 \qquad \forall\, i, j, \tag{23}$$

$$[a_i, a_j^\dagger] = 0 \qquad \forall\, i \neq j, \tag{24}$$

$$[a_i, a_i^\dagger] = 1 - 2n_i \qquad \text{with } n_i = c_i^\dagger c_i, \tag{25}$$

using that

$$\{Z_i, c_j^{(+)}\} = 0 \tag{26}$$

The corresponding proofs can be found in section 3.8.

3.5 The occupation-number basis in the new operators

The basis states of the fermionic Hilbert space can be written as

$$\begin{aligned}
|n_1, n_2, \ldots, n_L\rangle &\equiv \left(\prod_{i=1}^{L} (c_i^\dagger)^{n_i}\right) |0\rangle && \text{w. } n_i = 0, 1 \\
&= \left\{\prod_{i=1}^{L} \left[a_i^\dagger \left(\prod_{j=1}^{i-1} Z_j\right)\right]^{n_i}\right\} |0\rangle && (c_1^\dagger = a_1^\dagger) \\
&= (a_1^\dagger)^{n_1} (a_2^\dagger Z_1)^{n_2} (a_3^\dagger Z_2 Z_1)^{n_3} \cdots (a_L^\dagger Z_{L-1} \cdots Z_1)^{n_L} |0\rangle \\
&= \left(\prod_{i=1}^{L} (a_i^\dagger)^{n_i}\right) (Z_1)^{n_2} (Z_2 Z_1)^{n_3} \cdots (Z_{L-1} \cdots Z_1)^{n_L} |0\rangle \\
&= \left(\prod_{i=1}^{L} (a_i^\dagger)^{n_i}\right) |0\rangle && \text{w. } n_i = 0, 1,
\end{aligned} \tag{27}$$

using $[Z_i, Z_j] = 0\ \forall i, j$, $[Z_i, c_j] = 0$ for $i \neq j$ (hence $[Z_i, a_j^\dagger] = 0$ for $i \neq j$), and $Z_j |0\rangle = |0\rangle$. The ordering is not important anymore because $[a_i^\dagger, a_j^\dagger] = 0$.

3.6 Action on basis states

One cannot define the action of a_j ($a_j^\dagger$) locally, since the JW string $Z_1 \cdots Z_{j-1}$ is non-local, so one has to examine them on the full many-body basis. For $n_\ell = 0$,

$$
\begin{aligned}
a_\ell \left| n_1, \ldots, n_\ell = 0, \ldots, n_L \right\rangle &= a_\ell \left(\prod_{i=1}^{L} (a_i^\dagger)^{n_i} \right) |0\rangle \\
&\overset{[a_\ell, a_j^\dagger]=0}{=} \left(\prod_{i=1}^{L} (a_i^\dagger)^{n_i} \right) a_\ell |0\rangle = 0,
\end{aligned} \tag{28}
$$

since $a_\ell |0\rangle = c_\ell |0\rangle = 0$. For $n_\ell = 1$,

$$
\begin{aligned}
a_\ell \left| n_1, \ldots, n_\ell = 1, \ldots, n_L \right\rangle &= a_\ell \left(\prod_{\substack{i=1 \\ i \neq \ell}}^{L} (a_i^\dagger)^{n_i} \right) a_\ell^\dagger |0\rangle \\
&\overset{[a_\ell, a_j^\dagger]=0}{=} \left(\prod_{\substack{i=1 \\ i \neq \ell}}^{L} (a_i^\dagger)^{n_i} \right) a_\ell a_\ell^\dagger |0\rangle = \left(\prod_{\substack{i=1 \\ i \neq \ell}}^{L} (a_i^\dagger)^{n_i} \right) c_\ell c_\ell^\dagger |0\rangle \\
&= \left(\prod_{\substack{i=1 \\ i \neq \ell}}^{L} (a_i^\dagger)^{n_i} \right) |0\rangle = \left| n_1, \ldots, n_\ell = 0, \ldots, n_L \right\rangle .
\end{aligned} \tag{29}
$$

Using $[a_\ell^\dagger, a_j^\dagger] = 0$ and $(a_\ell^\dagger)^2 = 0$, one quickly finds

$$
a_\ell^\dagger \left| n_1, \ldots, n_\ell = 0, \ldots, n_L \right\rangle = \left| n_1, \ldots, n_\ell = 1, \ldots, n_L \right\rangle , \tag{30}
$$

$$
a_\ell^\dagger \left| n_1, \ldots, n_\ell = 1, \ldots, n_L \right\rangle = 0. \tag{31}
$$

Thus a_ℓ ($a_\ell^\dagger$) destroys (creates) a particle in the many-body state *without* giving rise to an extra sign structure (as c_ℓ ($c_\ell^\dagger$) would do).

3.7 Identification with spins

We can thus identify $n_i = 0, 1$ with $n_i = \downarrow, \uparrow$ in the many-body state, and the new operators can be associated with spins by identifying

$$
a_j = S_j^-, \qquad a_j^\dagger = S_j^+ \qquad \& \qquad n_j - \tfrac{1}{2} = S_j^z = \tfrac{1}{2} \sigma_j^z, \tag{32}
$$

since $[S_j^-, S_j^+] = -2 S_j^z$ for spins (see the quick refresher on spin, section 6). For instance

$$
a_3^\dagger |0\rangle = S_3^+ \left| \downarrow\downarrow\downarrow \right\rangle = \left| \uparrow\downarrow\downarrow \right\rangle .
$$

Matrix elements do not change under the JWT, since it is only a rewriting of operators and a renaming of basis states.

3.8 Proofs about the JWT

Claim: $\{Z_j, c_j^{(+)}\} = 0$**.**

Proof. On the single site j, with $Z_j |0\rangle_j = |0\rangle_j$, $Z_j |1\rangle_j = -|1\rangle_j$, $c_j |0\rangle_j = 0$ and $c_j |1\rangle_j = |0\rangle_j$,

$$Z_j c_j |0\rangle_j = Z_j \cdot 0 = 0 = c_j |0\rangle_j = c_j Z_j |0\rangle_j \,, \tag{33}$$
$$Z_j c_j |1\rangle_j = Z_j |0\rangle_j = |0\rangle_j = c_j |1\rangle_j \,, \tag{34}$$
$$c_j Z_j |1\rangle_j = -c_j |1\rangle_j = -|0\rangle_j \,. \tag{35}$$

Hence, on both basis states $\{Z_j, c_j\} = Z_j c_j + c_j Z_j = c_j - c_j = 0$, and since $Z_j^\dagger = Z_j$ it follows that $\{Z_j, c_j^\dagger\} = 0$ as well. (see Coleman, Chapter 4.1) $\square$

Claim: $[a_i^{(+)}, a_j^{(+)}] = 0 \ \ \forall\, i, j$**.**

Proof. Assume w.l.o.g. that $i < j$. Using eq. (22),

$$\begin{aligned} a_j a_i &= (Z_1 \cdots Z_{j-1} c_j)\,(Z_1 \cdots Z_{i-1} c_i) \\ &\overset{\substack{[Z_j, c_\ell]=0,\, j\neq\ell \\ \{c_i, c_j\}=0}}{=} Z_1^2 \cdots Z_{i-1}^2\, Z_i \cdots Z_{j-1}\, c_j c_i = -Z_i \cdots Z_{j-1}\, c_i c_j \qquad (= 0 \text{ if } i = j), \end{aligned} \tag{36}$$

and likewise

$$\begin{aligned} a_i a_j &= (Z_1 \cdots Z_{i-1} c_i)\,(Z_1 \cdots Z_{j-1} c_j) \\ &\overset{[Z_i, c_\ell]=0,\, i\neq\ell}{=} Z_{i+1} \cdots Z_{j-1}\, c_i Z_i c_j \overset{\{Z_i, c_i\}=0}{=} -Z_i \cdots Z_{j-1}\, c_i c_j. \end{aligned} \tag{37}$$

Comparing the two, $a_i a_j = a_j a_i$, i.e. $[a_i, a_j] = 0$. Analogously for $[a_i^\dagger, a_j^\dagger]$ and for $i > j$; the case $i = j$ is straightforward since $c_i^2 = 0$. $\square$

Claim: $[a_i, a_j^\dagger] = 0 \ \ \forall\, i \neq j$**.**

Proof. We can show it w.l.o.g. for $i < j$. Using eq. (22) and eq. (26),

$$\begin{aligned} a_i^\dagger a_j &= \big(c_i^\dagger Z_{i-1} \cdots Z_1\big)\big(Z_1 \cdots Z_{j-1} c_j\big) \overset{Z_i^2=1,\, i<j}{=} c_i^\dagger\, Z_i \cdots Z_{j-1}\, c_j \\ &= Z_{i+1} \cdots Z_{j-1}\, c_i^\dagger Z_i c_j \overset{\{Z_i, c_i^\dagger\}=0}{=} -Z_i \cdots Z_{j-1}\, c_i^\dagger c_j = +Z_i \cdots Z_{j-1}\, c_j c_i^\dagger, \end{aligned} \tag{38}$$

and

$$\begin{aligned} a_j a_i^\dagger &= \big(Z_1 \cdots Z_{j-1} c_j\big)\big(c_i^\dagger Z_{i-1} \cdots Z_1\big) \\ &= Z_i \cdots Z_{j-1}\, c_j c_i^\dagger\, Z_{i-1}^2 \cdots Z_1^2 \qquad ([Z_j, c_\ell^{(+)}] = 0,\ j \neq \ell) \\ &= Z_i \cdots Z_{j-1}\, c_j c_i^\dagger. \end{aligned} \tag{39}$$

Both reduce to the same expression, so $a_j a_i^\dagger - a_i^\dagger a_j = 0$ for $i < j$, i.e. $[a_j, a_i^\dagger] = 0$; the case $i > j$ follows straightforwardly. $\square$

Claim: $[a_i, a_i^\dagger] = 1 - 2n_i$ **with** $n_i = c_i^\dagger c_i$**.**

Proof. Using eq. (22) and $Z_k^2 = \mathbb{1}$,

$$a_i^\dagger a_i = c_i^\dagger \, Z_{i-1} \cdots Z_1 Z_1 \cdots Z_{i-1} \, c_i = c_i^\dagger c_i = n_i, \tag{40}$$

$$a_i a_i^\dagger = Z_1 \cdots Z_{i-1} \, c_i c_i^\dagger \, Z_{i-1} \cdots Z_1 = c_i c_i^\dagger = 1 - c_i^\dagger c_i, \tag{41}$$

so that

$$[\hat{a}_i, \hat{a}_i^\dagger] = a_i a_i^\dagger - a_i^\dagger a_i = (1 - c_i^\dagger c_i) - c_i^\dagger c_i = 1 - 2\hat{n}_i. \tag{42}$$

□

4 Minimizing complex functions with Wirtinger derivatives

4.1 Definition

Define for $z = x + iy \in \mathbb{C}$ with $x, y \in \mathbb{R}$

$$\frac{\partial}{\partial z} \equiv \frac{1}{2}\Big(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\Big) \qquad \& \qquad \frac{\partial}{\partial \bar{z}} = \frac{1}{2}\Big(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\Big) \tag{43}$$

for functions of one complex variable, and

$$\frac{\partial}{\partial z_j} = \frac{1}{2}\Big(\frac{\partial}{\partial x_j} - i\frac{\partial}{\partial y_j}\Big) \qquad \& \qquad \frac{\partial}{\partial \bar{z}_j} = \frac{1}{2}\Big(\frac{\partial}{\partial x_j} + i\frac{\partial}{\partial y_j}\Big) \tag{44}$$

for functions of multiple complex variables.

4.2 Agreement with the complex derivative

If f is complex differentiable at a point (i.e. the limit $f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$ exists), then $\frac{\partial f}{\partial z}$ agrees with the complex derivative $\frac{df}{dz}$.

Proof. For $f(z) = u(z) + iv(z)$ complex differentiable,

$$\begin{aligned} \frac{\partial f}{\partial z} &\overset{(43)}{=} \frac{1}{2}\Big(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}\Big) = \frac{1}{2}\Big(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} - i\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}\Big) \\ &\overset{(43)}{=} \frac{\partial u}{\partial z} + i\frac{\partial v}{\partial z} \overset{\text{Def.}}{=} \frac{df}{dz}. \end{aligned} \tag{45}$$

$\square$

4.3 The $\bar{z}$-derivative and Cauchy–Riemann

Furthermore,

$$\begin{aligned} 0 \overset{!}{=} \frac{\partial f}{\partial \bar{z}} &= \frac{1}{2}\Big(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\Big) = \frac{1}{2}\Big(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} + i\frac{\partial u}{\partial y} - \frac{\partial v}{\partial y}\Big) \\ &\iff \quad \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \wedge \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}, \end{aligned} \tag{46}$$

so that $0 = \dfrac{\partial f}{\partial \bar{z}}$ is equivalent to the Cauchy–Riemann equations, i.e. the necessary & sufficient condition for f to be complex differentiable.

4.4 Minimizing a real-valued function

Minimize a real-valued function[1]

$$f\colon \mathbb{C} \to \mathbb{R} \tag{47}$$

$$f(z) = f(x, y), \qquad (z = x + iy) \tag{48}$$

via the condition

$$\nabla f(x, y) = \begin{pmatrix} \partial_x f \\ \partial_y f \end{pmatrix} \stackrel{!}{=} 0. \tag{49}$$

Then

$$\partial_x f \stackrel{!}{=} 0 \stackrel{!}{=} \partial_y f \quad \iff \quad \frac{\partial f}{\partial \bar{z}} = \frac{1}{2}\Big(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\Big) \stackrel{!}{=} 0, \tag{50}$$

so one can minimize a real-valued function by imposing $\boxed{\dfrac{\partial f}{\partial \bar{z}} = 0}$.

[1] Minimizing a complex-valued f does not make sense because $\mathbb{C}$ has no order.

5 The density matrix, entanglement entropy and their connection to singular value decomposition (SVD)

5.1 Main goal

Prove that for λ_i the eigenvalues of the reduced density matrix and s_i the Schmidt values,

$$\boxed{S_{\text{Neum}} \equiv \text{Tr}\big(\rho_A \log \rho_A\big) = \sum_i \lambda_i \log \lambda_i = \sum_i s_i^2 \log s_i^2}. \tag{51}$$

5.2 The density operator

Let $\{p_i, |\psi_i\rangle\}$ be an ensemble of pure states, describing a system that is in one of the states $|\psi_i\rangle$ with probability p_i, i.e. the system is not fully known. Define the density operator as

$$\rho = \sum_i p_i \, |\psi_i\rangle\langle\psi_i| \,, \qquad p_i \in \mathbb{R} \;\; \& \;\; \sum_i p_i = 1, \tag{52}$$

which is a positive semi-definite operator of trace one (see for example Nielsen & Chuang).

5.3 Observables on a composite system

Take a system A with some observable M, and a system B, giving the composite system AB (see Box 2.6). Assume we can measure M on system A; then a corresponding measurement on AB is $\tilde{M} = M \otimes \mathbb{1}_B$.

Proof. Take AB to be prepared in $|m\rangle\,|\psi\rangle$ with $M\,|m\rangle = m\,|m\rangle$. Applying the measurement to $|m\rangle\,|\psi\rangle$ should yield m with probability one, i.e. the corresponding projector for $\tilde{M}$ is $P_m \otimes \mathbb{1}_B$. By the spectral theorem, $\tilde{M} = \sum_m m\, P_m \otimes \mathbb{1}_B = M \otimes \mathbb{1}_B$ for a general state of AB. □

5.4 The reduced density matrix as a partial trace

Measurement averages must be the same whether computed via the full ρ^{AB} or via some ρ^A associated to system A, i.e.

$$\text{tr}\big(M\rho^A\big) \stackrel{!}{=} \text{tr}\big(\tilde{M}\rho^{AB}\big) = \text{tr}\big((M \otimes \mathbb{1}_B)\rho^{AB}\big). \tag{53}$$

Take $\rho^A = \text{tr}_B(\rho^{AB})$ where

$$\text{tr}_B\big(|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|\big) \equiv |a_1\rangle\langle a_2| \; \text{tr}\big(|b_1\rangle\langle b_2|\big) \qquad \& \qquad \text{tr}_B(\cdot) \text{ linear}. \tag{54}$$

Write ρ^{AB} by using the eigenbasis of M for system A with $M\,|i\rangle = m_i\,|i\rangle$ and $\langle i|j\rangle = \delta_{ij}$:

$$\rho^{AB} = \sum_\alpha p^\alpha \, |\psi\rangle\langle\psi|^\alpha = \sum_\alpha p^\alpha \sum_{ijk\ell} c^\alpha_{ij} (c^\alpha_{k\ell})^* \; |i\rangle_A \, |j\rangle_B \, \langle\ell|_B \, \langle k|_A \,, \tag{55}$$

$$\rho^A = \sum_\alpha \sum_{ijk\ell} p^\alpha c^\alpha_{ij} (c^\alpha_{k\ell})^* \; |i\rangle\langle k|_A \, \underbrace{\text{tr}\big(|j\rangle\langle\ell|_B\big)}_{=\delta_{j\ell}} = \sum_{\alpha,ijk} p^\alpha c^\alpha_{ij} (c^\alpha_{kj})^* \; |i\rangle\langle k|_A \,. \tag{56}$$

Therefore

$$\operatorname{tr}\big(M\rho^A\big) = \sum_{\alpha,ijk} p^\alpha c^\alpha_{ij}(c^\alpha_{kj})^* \underbrace{\operatorname{tr}\big(m_i\,|i\rangle\langle k|_A\big)}_{=m_i\delta_{ik}} = \sum_{\alpha,ij} p^\alpha m_i\, c^\alpha_{ij}(c^\alpha_{ij})^*, \tag{57}$$

and similarly $\operatorname{tr}\big((M\otimes \mathbb{1}_B)\rho^{AB}\big) = \sum_{\alpha,ij} p^\alpha m_i\, c^\alpha_{ij}(c^\alpha_{ij})^*$, which proves the claim. One can also easily show that this is the only map that conserves consistency (see Box 2.6).

5.5 Reduced density matrix of a pure state

Now examine the reduced density matrix for a pure state,

$$\rho_{AB} = |\psi\rangle\langle\psi| = \sum_{ijk\ell} c_{ij}(c_{k\ell})^*\ |i\rangle_A\,|j\rangle_B\,\langle\ell|_B\,\langle k|_A\,, \tag{58}$$

$$\rho_A = \operatorname{tr}_B(\rho_{AB}) = \sum_{ijk} c_{ij}(c_{kj})^*\ |i\rangle\langle k|_A\,, \tag{59}$$

such that the matrix representation is

$$\boxed{\rho_A = \psi\psi^\dagger}, \tag{60}$$

using that the c_{ij} parametrize the matrix $\psi \in \mathbb{C}^{d_A\times d_B}$, i.e. $|\psi\rangle = \sum_{ij} c_{ij}\,|i\rangle_A\,|j\rangle_B$.

5.6 Connection to the SVD

Now use the SVD on ψ:

$$\psi = USV^\dagger \qquad \text{with } U\in\mathbb{C}^{d_A\times d_A},\ V\in\mathbb{C}^{d_B\times d_B} \text{ unitary.} \tag{61}$$

Then, using $V^\dagger V = \mathbb{1}$,

$$\rho_A = USV^\dagger VSU^\dagger = US^2U^\dagger, \qquad S^2 = \operatorname{diag}(s_1^2, s_2^2, \dots) \in \mathbb{R}^{d_A\times d_A}_{\geq 0}. \tag{62}$$

Furthermore, since ρ_A is hermitian we can diagonalize it with $\rho_A = ODO^\dagger = US^2U^\dagger$ and $D = \operatorname{diag}(\lambda_1,\dots)$, so that

$$\boxed{\lambda_i = s_i^2}, \tag{63}$$

i.e. the eigenvalues are given by the (squared) Schmidt values.

5.7 Functions of the density matrix via the spectral theorem

Since ρ_A is hermitian, we can use the spectral theorem to write it as $\rho_A = \sum_i \lambda_i P_i$ where λ_i are its eigenvalues and the eigenvectors $|u_i\rangle$ give $P_i = |u_i\rangle\langle u_i|$, since

$$\rho_A = UDU^\dagger = \big(|u_1\rangle\,,\dots,|u_n\rangle\big)\begin{pmatrix}\lambda_1 & & \\ & \ddots & \\ & & \lambda_n\end{pmatrix}\begin{pmatrix}\langle u_1| \\ \vdots \\ \langle u_n|\end{pmatrix}, \qquad \langle u_i|u_j\rangle = \delta_{ij}. \tag{64}$$

Thus a function $f(\cdot)$ applied to ρ_A gives

$$
\begin{aligned}
f(\rho_A) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \rho_A^n \overset{(*)}{=} \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \sum_i \lambda_i^n \left|u_i\right\rangle\!\left\langle u_i\right| \\
&= \sum_i \Big(\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \lambda_i^n \Big) \left|u_i\right\rangle\!\left\langle u_i\right| = \sum_i f(\lambda_i) \left|u_i\right\rangle\!\left\langle u_i\right| = U \operatorname{diag}\big(f(\lambda_i)\big) U^\dagger,
\end{aligned}
\tag{65}
$$

where $(*)$ uses $\rho_A^n = \sum_i \lambda_i^n \left|u_i\right\rangle\!\left\langle u_i\right|$ (e.g. $\rho_A \rho_A = \sum_i \lambda_i^2 \left|u_i\right\rangle\!\left\langle u_i\right|$).

5.8 The Von Neumann entanglement entropy

Therefore, for the Von Neumann entanglement entropy,

$$
\begin{aligned}
S = \operatorname{Tr}\big(\rho_A \log \rho_A\big) &= \operatorname{Tr}\big(\rho_A\, U \log(\operatorname{diag}(\lambda_i))\, U^\dagger\big) \\
&= \operatorname{Tr}\big(U \operatorname{diag}(\lambda_i) \underbrace{U^\dagger U}_{=\mathbb{1}} \log(\operatorname{diag}(\lambda_i)) U^\dagger\big) \\
&= \sum_i \lambda_i \log \lambda_i = \boxed{\sum_i s_i^2 \log s_i^2}.
\end{aligned}
\tag{66}
$$

S is zero for two unentangled systems and non-zero for entangled systems.

5.9 Example

In our case, for example, $L = 2$, $N_f = 1$, $N_{\mathrm{ph}} = 1$:

$$\{\left|01\right\rangle, \left|10\right\rangle\} \;\&\; \{\left|0\right\rangle, \left|1\right\rangle\} \quad \text{as bases of } \mathcal{H}_f \;\&\; \mathcal{H}_{\mathrm{ph}}, \tag{67}$$

$$\Rightarrow \{\left|01, 0\right\rangle;\ \left|01, 1\right\rangle;\ \left|10, 0\right\rangle;\ \left|10, 1\right\rangle\} \quad \text{as basis of } \mathcal{H}_f \otimes \mathcal{H}_{\mathrm{ph}}, \tag{68}$$

$$\Rightarrow \left|\psi\right\rangle = \sum_{i=1}^{2} \sum_{j=1}^{2} \psi_{ij} \left|i\right\rangle_f \left|j\right\rangle \quad \text{with } (\psi)_{ij} = \psi_{ij} \;\&\; \psi \in \mathbb{C}^{2\times 2}, \tag{69}$$

$$\Rightarrow S(\rho_A) = \sum_\ell s_\ell^2 \log s_\ell^2. \tag{70}$$

Note that since $\operatorname{tr} \rho^{AB} = \operatorname{tr} \rho^A = \operatorname{tr} \rho^B = 1$ we must have $\sum_\ell \lambda_\ell = \sum_\ell s_\ell^2 = 1$.

5.10 References

- M. Nielsen & I. Chuang, *Quantum Computation and Quantum Information* (density operator, partial trace; Box 2.6).
- U. Schollwöck, *The density-matrix renormalization group in the age of matrix product states*, beginning of Chapter 4 (MPS).

6 Quick refresher on spin

Spin can be described with the operators $\hat{S}_x, \hat{S}_y, \hat{S}_z$ that obey the following commutation relations:

$$\boxed{[\hat{S}_j, \hat{S}_k] = i\hbar\, \epsilon_{jk\ell} \hat{S}_\ell} \qquad (\hbar = 1), \tag{71}$$

with $i, j, k = x, y, z = 1, 2, 3$ and using the Einstein summation convention. One can define raising and lowering operators as

$$\boxed{\hat{S}_\pm = \hat{S}_x \pm i\hat{S}_y} \qquad \text{with} \qquad \boxed{[S_-, S_+] = -2S_z}. \tag{72}$$

Proof.

$$\begin{aligned} [S_-, S_+] &= [S_x - iS_y,\ S_x + iS_y] \\ &= [S_x, S_x] + i[S_x, S_y] - i[S_y, S_x] + [S_y, S_y] \\ &= 2i[S_x, S_y] \overset{(71)}{=} 2i \cdot i\, \epsilon_{12\ell} S_\ell = -2\, \epsilon_{12\ell} S_\ell = -2S_z, \end{aligned} \tag{73}$$

where the first step used the bilinearity of the commutator and the last step uses $\epsilon_{jk\ell} = 0$ if two indices are equal, leaving $\epsilon_{123} S_3 = S_z$. □

For spin-$\frac{1}{2}$ particles, one can write

$$S_j = \frac{\hbar}{2}\, \sigma_j \qquad \text{with } \sigma_j \text{ the Pauli matrices, } j = x, y, z, \tag{74}$$

and often again $\hbar = 1$.

7 Quick refresher on unitary transformations

Let's say we have some Hamiltonian $\hat{H}$, which we examine in a basis $\{|v_j\rangle\}$ with $j = 1, \dots, \dim(\mathcal{H})$ and $\hat{H} \colon \mathcal{H} \to \mathcal{H}$. The matrix representation of the operator $\hat{H}$ in this basis is thus given by

$$H_{ij} = \left\langle v_i \middle| \hat{H}\, v_j \right\rangle = \langle v_i| \hat{H} |v_j\rangle\,. \tag{75}$$

Applying a unitary transformation U (with $U^\dagger U = \mathbb{1} = UU^\dagger$), i.e.

$$|v_j\rangle \to |\tilde{v}_j\rangle = U\,|v_j\rangle \qquad \& \qquad \hat{H} \to \hat{\tilde{H}} = U\hat{H}U^\dagger, \tag{76}$$

yields the matrix elements

$$\tilde{H}_{ij} = \langle \tilde{v}_i| \hat{\tilde{H}} |\tilde{v}_j\rangle = \langle v_i| \underbrace{U^\dagger U}_{=\mathbb{1}} \hat{H} \underbrace{U^\dagger U}_{=\mathbb{1}} |v_j\rangle = \langle v_i| \hat{H} |v_j\rangle = H_{ij}. \tag{77}$$

Hence the matrix representation of an operator does *not* change under a unitary transformation.

8 Quick refresher on the current in electrodynamics

8.1 Maxwell's equations

Given Maxwell's equations,

$$\vec{\nabla} \cdot \vec{E}(\vec{r}, t) = \frac{1}{\varepsilon_0}\, \rho(\vec{r}, t) \tag{78}$$

$$\vec{\nabla} \cdot \vec{B}(\vec{r}, t) = 0 \tag{79}$$

$$\vec{\nabla} \times \vec{E}(\vec{r}, t) = -\frac{\partial}{\partial t} \vec{B}(\vec{r}, t) \tag{80}$$

$$\vec{\nabla} \times \vec{B}(\vec{r}, t) = \frac{1}{c^2} \frac{\partial}{\partial t} \vec{E}(\vec{r}, t) + \frac{1}{\varepsilon_0 c^2}\, \vec{j}(\vec{r}, t) \tag{81}$$

8.2 Continuity equation

From $\varepsilon_0\, \partial_t$ (*eq.* (78)), using $\partial_x \partial_t = \partial_t \partial_x$,

$$\frac{\partial}{\partial t} \rho(\vec{r}, t) = \varepsilon_0\, \vec{\nabla} \cdot \big(\partial_t \vec{E}(\vec{r}, t)\big), \tag{82}$$

and from $\varepsilon_0 c^2\, \vec{\nabla} \cdot$ (*eq.* (81)),

$$\begin{aligned} \vec{\nabla} \cdot \vec{j}(\vec{r}, t) &= \vec{\nabla} \cdot \big(\vec{\nabla} \times \vec{B}(\vec{r}, t)\big) - \varepsilon_0\, \vec{\nabla} \cdot \big(\partial_t \vec{E}(\vec{r}, t)\big) \\ &= -\varepsilon_0\, \vec{\nabla} \cdot \big(\partial_t \vec{E}(\vec{r}, t)\big) \qquad (\mathrm{div}(\mathrm{rot}\, \vec{F}) = 0). \end{aligned} \tag{83}$$

Adding the two gives the continuity equation

$$\frac{\partial}{\partial t} \rho(\vec{r}, t) + \vec{\nabla} \cdot \vec{j}(\vec{r}, t) = 0. \tag{84}$$

8.3 Point-particle solutions

Solutions of the continuity equation:

$$\rho(\vec{r}, t) = \sum_\alpha q_\alpha\, \delta\big(\vec{r} - \vec{r}_\alpha(t)\big) \tag{85}$$

$$\vec{j}(\vec{r}, t) = \sum_\alpha q_\alpha\, \vec{v}_\alpha(t)\, \delta\big(\vec{r} - \vec{r}_\alpha(t)\big) \tag{86}$$

Proof. Note that

$$\begin{aligned} \partial_t\, \delta\big(\vec{r} - \vec{r}_\alpha(t)\big) &= \partial_t \Big(\delta\big(r_x - r_x^\alpha(t)\big) \delta\big(r_y - r_y^\alpha(t)\big) \delta\big(r_z - r_z^\alpha(t)\big)\Big) \\ &= -\sum_{i=x,y,z} \dot{r}_i^\alpha(t)\, \partial_{r_i} \delta\big(r_i - r_i^\alpha(t)\big) = -\dot{\vec{r}}_\alpha(t) \cdot \vec{\nabla}\, \delta\big(\vec{r} - \vec{r}_\alpha(t)\big), \end{aligned} \tag{87}$$

where we used the chain rule in the second line and

$$\vec{\nabla} \cdot \Big(\vec{v}_\alpha(t)\, \delta\big(\vec{r} - \vec{r}_\alpha(t)\big)\Big) = \sum_{i=x,y,z} \partial_{r_i} \Big(v_i^\alpha(t)\, \delta\big(\vec{r} - \vec{r}_\alpha(t)\big)\Big)$$
$$= \sum_{i=x,y,z} v_i^\alpha(t)\, \partial_{r_i} \delta\big(\vec{r} - \vec{r}_\alpha(t)\big) = \vec{v}_\alpha(t) \cdot \vec{\nabla}\, \delta\big(\vec{r} - \vec{r}_\alpha(t)\big). \tag{88}$$

Therefore

$$\frac{\partial}{\partial t} \rho(\vec{r}, t) = -\sum_\alpha q_\alpha\, \dot{\vec{r}}_\alpha(t) \cdot \Big(\vec{\nabla}\, \delta\big(\vec{r} - \vec{r}_\alpha(t)\big)\Big) = -\vec{\nabla} \cdot \vec{j}(\vec{r}, t), \tag{89}$$

which is (84). $\square$

Note that these solutions are not unique (one equation for four components).